

Secure Blockchain Voting System - Updated Technical Documentation

Project Name: VoteChain

Project Type: Private Blockchain Voting System with Fingerprint Authentication, Ephemeral Hashes, and Multi-Node Consensus

Target Users: ~1,000 students

Timeline: 17 Sept 2025 → 13 Feb 2026

Development Team Roles: - **Core Developers (You + Rajib):** Blockchain, consensus, ephemeral hash, fingerprint integration in C++ - **Frontend Developer (Abhash):** UI/UX, React.js/Next.js, registration & voting pages - **Backend Developer (Sandesh):** Node.js + Express, API endpoints, database, blockchain integration - **Testing (Ayush):** Unit testing, load testing, security checks, multi-node testing

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1. Introduction

Overview

VoteChain is a decentralized, privacy-preserving voting platform designed for secure elections. By leveraging blockchain, fingerprint authentication, and ephemeral hash techniques, it ensures voter anonymity, fraud prevention, and tamper-proof vote storage.

Key Features

- **Decentralized & Immutable:** Votes cannot be altered once recorded on blockchain.
- **Voter Privacy:** No PII is stored; ephemeral hashes unlink votes after election.
- **Fraud Prevention:** Duplicate votes are flagged and moved to Dead Block.
- **Auditability:** Votes are verifiable while keeping voter identity hidden.
- **Scalability:** Supports 1K+ users via efficient hash-based identity management.

System Design Principles

- **Security First:** End-to-end encryption, ephemeral hashes, and digital signatures.
 - **Decentralization:** Multi-node blockchain without central authority.
 - **Transparency with Anonymity:** Votes are auditable, identities remain hidden.
 - **User-Friendly Design:** Multi-device support with accessible UI.
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2. Voter Authentication

Key Generation Process

1. Voter provides college ID.
2. Fingerprint captured using scanner.
3. Permanent hash generated: `permanentHash = SHA256(voterID + fingerprintHash)`
4. Stored securely in backend.

Ephemeral Election-Based Hashing

1. Before each election, temporary hash generated: `tempHash = SHA256(permanentHash + electionName + electionDate + randomSalt)`

2. Temp hashes are election-specific and time-limited.
3. Discarded post-election to prevent linking to permanent identity.

Fingerprint Integration

- Fingerprint hash used to generate ephemeral hash for election.
 - Ensures one-person-one-vote while maintaining anonymity.
 - Validated before vote submission.
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3. Voting Process

Vote Submission

- Voter submits vote using ephemeral temp hash.
- Digitally signed to prevent forgery.

Blockchain Encryption

- Votes encrypted with asymmetric cryptography (ECC/AES-256).
- Stored on blockchain with timestamp.

Valid Vote Recognition

- Only first vote per temp hash accepted.
 - Subsequent votes flagged and moved to Dead Block.
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4. Duplicate Vote Handling

Dead Block Storage Mechanism

- Duplicate votes stored in Dead Block for manual review.

Role of Authorizers

- Can only see metadata (voter hash, timestamp).
- Can approve one vote or discard both.
- Any attempt to view vote data triggers automatic deletion.

Privacy-Preserving Verification

- Ensures voter anonymity during duplicate vote verification.
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5. Dead Block & Sybil Attack Protection

Sybil Attack Prevention

- Detects multiple fake voter attempts.
- Logged in Dead Block for auditing.

Metadata-Based Authorization

- Authorizers only access metadata to prevent bias.

Automatic Privacy Enforcement

- Unauthorized access attempts result in vote deletion.
 - Protects voter privacy at all levels.
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6. Limited-Time Voting & Post-Election Data Deletion

Data Expiry Mechanism

- Temp hashes and ephemeral data deleted post-election.

Tamper-Proof Record Keeping

- Blockchain keeps only authorized votes and invalid/fraud attempts.

Blockchain Validation

- Final vote count includes only verified transactions.
 - Ensures integrity and privacy.
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7. Security Architecture

Cryptographic Measures

- ECC for key generation.
- SHA-256 for permanent and ephemeral hash.
- AES-256 encryption for vote content.

Zero-Knowledge Proofs

- Prove voter eligibility without revealing identity.

Blockchain Consensus

- PoA or PBFT used for multi-node agreement.
 - Eliminates central authority and ensures immutability.
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8. System Scalability & Performance

- Hash-based identity reduces storage and computation.
 - Decentralized node network prevents single point of failure.
 - Load balancing ensures smooth voting for 1K+ users.
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9. User Accessibility & Voter Experience

- Simple, intuitive UI.
 - Supports web and mobile devices.
 - Assistive technologies for accessibility.
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10. Transparency & Auditability

- Votes publicly verifiable without exposing identity.
 - Fraud attempts logged but do not affect results.
 - Regulatory compliance maintained.
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11. Conclusion

VoteChain delivers a secure, anonymous, and verifiable voting experience using blockchain, ephemeral hashing, and biometric verification.

Future Enhancements

- AI-based fraud detection.
- Remote/cross-border voting support.
- Layer 2 scaling for faster transactions.

Ethical Considerations

- Data privacy compliance.
- Ensures equitable access to all voters.
- Prevents manipulation and maintains trust in the system.